



UNIVERSITY OF
OXFORD

Anomalous morphology of the inferior frontal cortex in children with developmental language disorder (DLD)

Nilgoun Bahar¹, Saloni Krishnan², Salomi S. Asaridou¹, Gabriel J. Cler³, & Kate E. Watkins¹

¹ Dept. of Experimental Psychology & Wellcome Trust Centre for Integrative Neuroimaging, University of Oxford, Oxford, UK; ² Dept. of Psychology, Royal Holloway, University of London, Surrey, UK; ³ Dept. of Speech & Hearing Sciences, University of Washington, Seattle, USA



INTRODUCTION

- Children with developmental language disorder (DLD) have persistent difficulties in learning their native language without a known cause
- We have limited knowledge of the brain basis and neurobiological risk factors associated with DLD
- Reported morphological abnormalities in DLD including hemispheric asymmetries are varied, based on small samples, using different methods
- As part of the Brain Organisation in Language Development (BOLD) study, we obtained structural MRI scans in a large, well-characterised sample of children aged 10-16 years with a range of language learning abilities

RESEARCH QUESTIONS

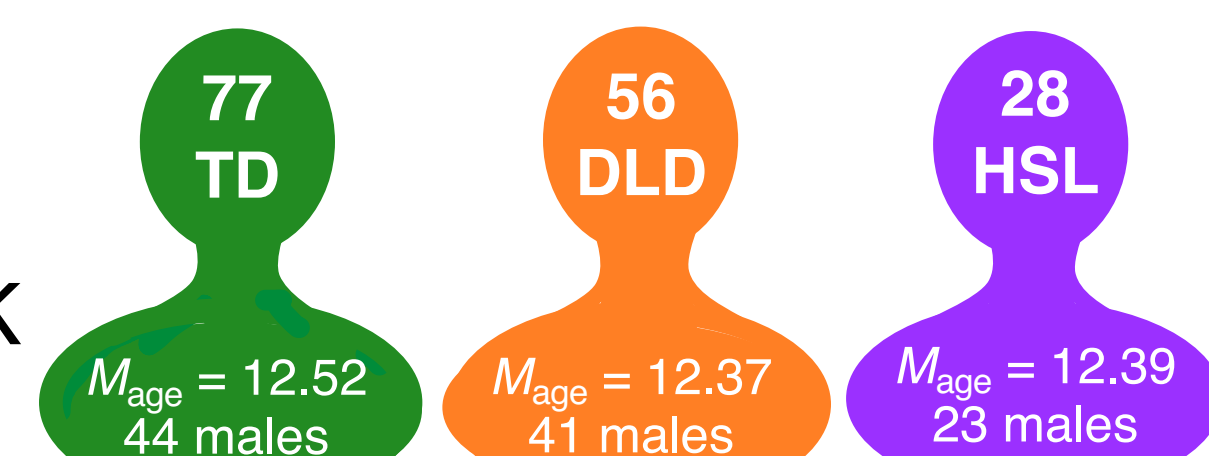
- Are there differences in cortical thickness and surface area in children with DLD relative to typically developing (TD) children?
- Are these differences related to performance on behavioural tests?
- Do children with DLD and TD children differ in terms of cortical asymmetry of perisylvian regions?

GROUP DEFINITION

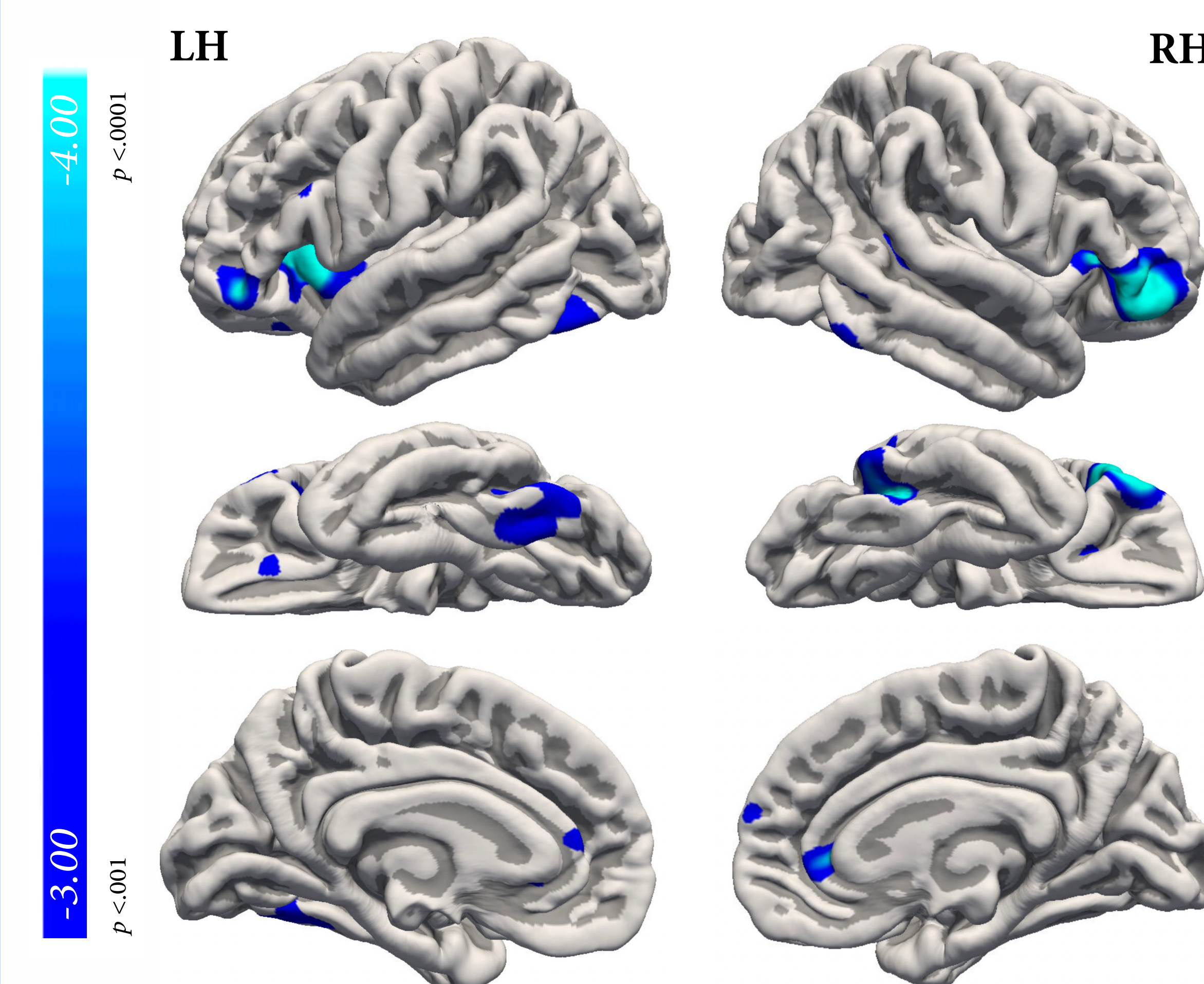
Group	Nonverbal IQ > 70	2+ language tests failed	History of speech & language problems
TD	✓		
DLD	✓	✓	✓
HSL	✓		✓

PARTICIPANTS

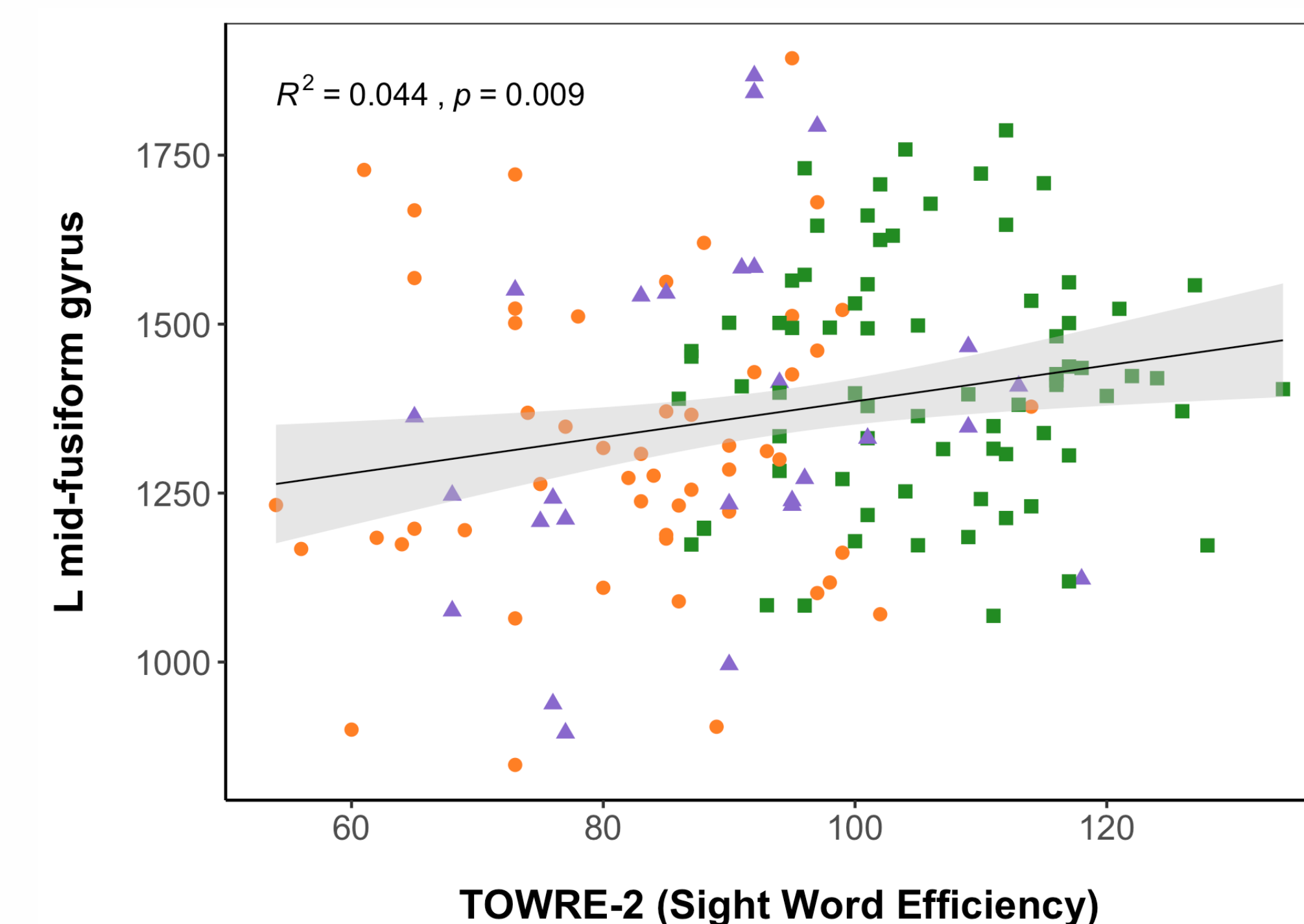
- N = 162 children
- 10-16 years old
- Grown up in the UK speaking English



RESULTS

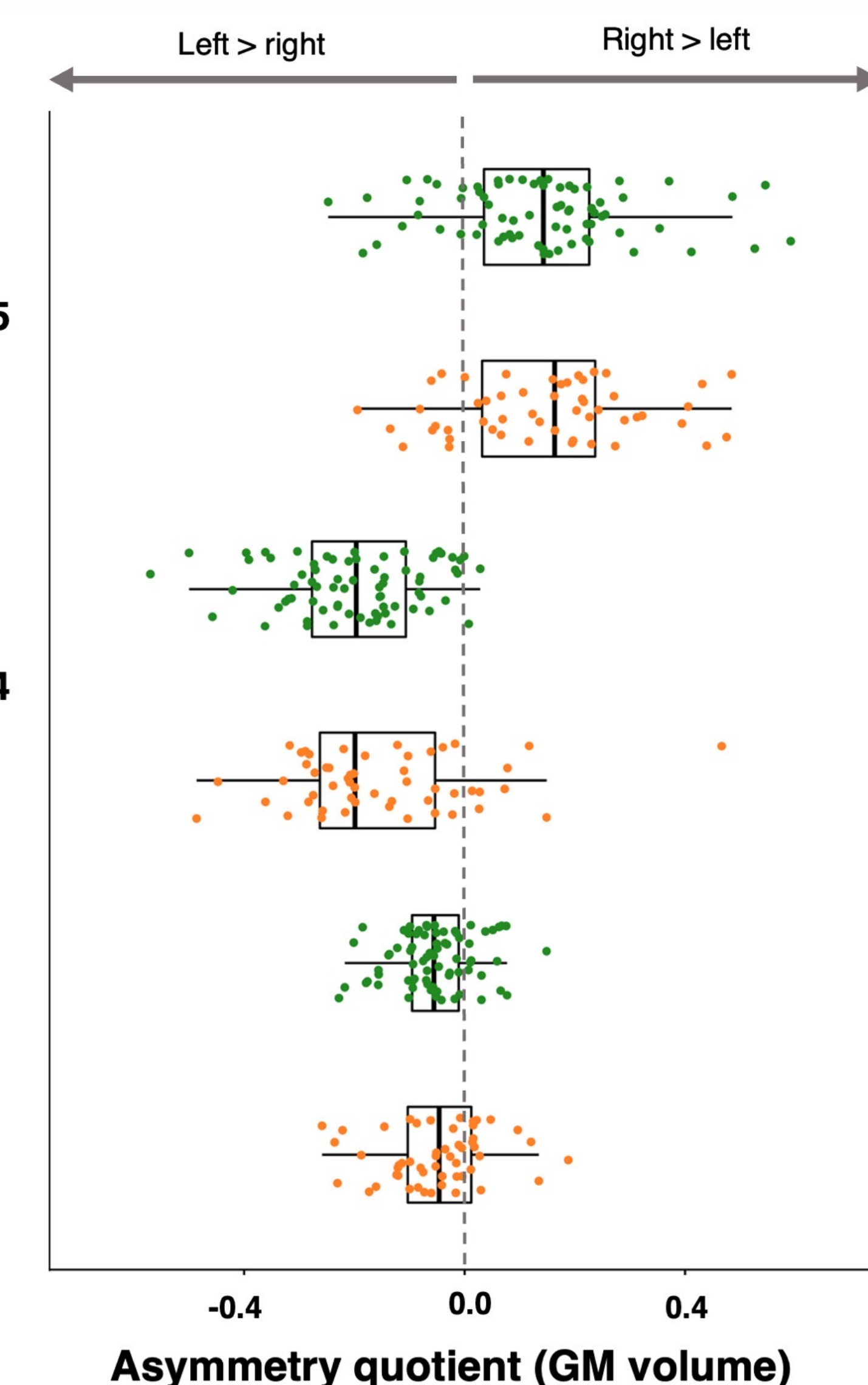


- Final participant numbers (73 TD, 52 DLD, 27 HSL)
- Lower surface area in DLD bilaterally in anterior IFG (pars orbitalis), extending into the anterior insula. Posteriorly, lower surface area in ventral occipito-temporal cortex (mid fusiform gyrus) bilaterally
- No differences between groups in cortical thickness
- Volume was also lower in DLD in the regions showing reduced surface area but only one cluster (right pars triangularis) survived correction for multiple comparisons



- Surface area in the left ventral occipito-temporal cortex correlated with word reading ability across the entire cohort of participants (including HSL)
- No correlation between lateral orbitofrontal region and measures of nonword repetition or oromotor praxis
- Asymmetry did not differ between DLD and TD in pars triangularis (BA45), pars opercularis (BA44) or STG (with or without controlling for sex, age, and NVIQ)
- Pars opercularis (BA44) and STG showed significant leftwards asymmetry, whereas pars triangularis (BA45) showed a significant rightward asymmetry in both groups

Orange: DLD; Green: TD



CONTACT & ACKNOWLEDGEMENTS

nilgoun.bahar@psy.ox.ac.uk
<https://boldstudy.wordpress.com/>



MATERIALS & METHODS

Behavioural measures

- Expressive and receptive vocabulary, grammar, and narrative; nonverbal IQ; verbal working memory and auditory verbal learning; nonword repetition and oromotor praxis; word and nonword reading skills

MRI acquisition

- Prisma 3T; 3D T1W MP-RAGE sequence

Image processing

- FreeSurfer group analysis pipeline
- Used Qoala-T for quality control of parcellated T1-weighted images (N=10 Rejected)
- Surface area (SA), cortical thickness (CT), and grey matter volume (GMV)

Whole-brain cortical morphometry

- GLM used to compare groups of DLD and TD in vertex-wise surface-based whole-brain analysis
- Outcome variables: SA, CT, GMV
- Covariates of no interest: sex, age
- Corrected for multiple comparisons with a cluster-wise threshold of $p < 0.05$ (two-tailed)

Asymmetry analysis

- Asymmetry quotient (AQ) for GMV in pars opercularis (BA44), pars triangularis (BA45), and superior temporal gyrus (STG) calculated using the following formula:

$$AQ = \frac{\text{Right} - \text{Left}}{0.5 \times (\text{Right} + \text{Left})}$$

CONCLUSIONS

- Large samples of children with DLD are necessary to identify statistically robust differences in cortical morphometry
- Novel findings of reduced surface area in inferior frontal cortex and ventral occipito-temporal cortex bilaterally in children with DLD; cortical thickness unaffected
- Surface area in left ventral occipito-temporal cortex related to reading performance
- Failed to replicate previous findings of reduced rightwards asymmetry in BA45 (pars triangularis) in children with DLD